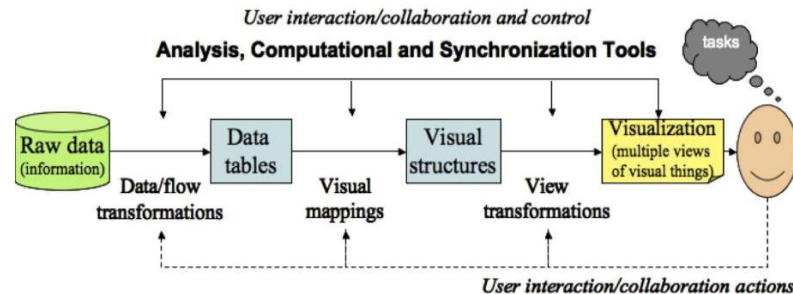


Unit-II

2.1 Foundations for Visualization: The end of the visualization pipeline namely getting data into the computer and on the human side how perception and cognition help us interpret images. We have looked at one fundamental visualization namely the scatterplot. There are many other visualization techniques and systems. To make sense of them we need to organize methods into categories or taxonomies. This is necessary for us to structure our study of the field. We first review the visualization pipeline and then discuss various ways to view the multitudes of techniques and systems that have been developed to date.

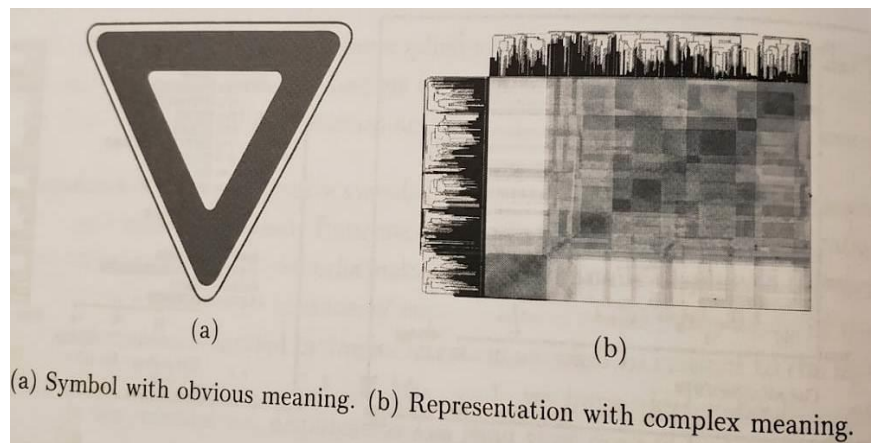


2.1.1 Visualization stages

The steps that need to be taken to define a visualization of data. Most visualization pipelines and systems map easily to these stages. Any transformation or computation can be placed at any of the stages. We selected this as the simplest representation of the process. We also note two key points: user interaction ideally takes place at any point in this pipeline (nodes and links) and each link is a many-to-many mapping. For example many visualization systems have multiple visualizations at the same time on the screen and thus have multiple representation mappings and corresponding renderings.

- **Data preprocessing and transformation:** The starting point is to process the raw data into something usable by the visualization system. The first part is to make sure that the data are mapped to fundamental data types for computer ingestion. The second step entails dealing with specific application data issues such as missing values, errors in input, and data too large for processing. The data may be simulated or sampled.
- **Mapping for visualizations:** Once the data are clean, we can decide on a specific visual representation. This requires representation mappings: geometry, color, and sound, for example. It is easy to simply develop nonsense visualization, or one that conveys the wrong information.
- **Rendering transformations:** The final stage involves mapping from geometry data to the image. This includes interfacing with a computer graphics Application Programmer's Interface (API). We need to select the viewing parameters shading technique if 3D device transformations. This stage of the pipeline is very dependent on the underlying graphics library.
- **Expressiveness:** An expressive visualization presents all the information, and only the information Expressiveness thus measures the concentration of information. Given information that we actually display to the user.
- **Effectiveness:** Visualization is effective when it can be interpreted accurately and quickly and when it can be rendered in a cost-effective manner. Effectiveness thus measures a specific cost of information perception.

2.1.2 Semiology of Graphical Symbols visual object called a graphical symbol. They often make up parts of visualizations (arrows, labels). We will look at how a graphical object or representation can be well designed and how it is perceived. The science of graphical symbols and marks is called semiology. Every possible construction in the Euclidean plane is a graphical representation made up of graphical symbols. This includes diagrams, networks, maps, plots, and other common visualizations. Semiology uses the qualities of the plane and objects on the plane to produce similarity features ordering features and proportionality



(a) Symbol with obvious meaning. (b) Representation with complex meaning.

features of the data that are visible for human consumption. There are numerous characteristics of visualizations of images or of graphics made up of symbols. We describe some of these in the following sections.

i. Symbols and Visualizations:

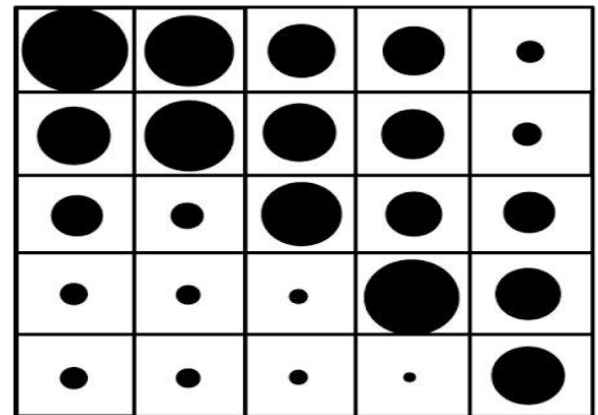
an (a) image that is universally recognizable (yield sign). Such images become preattentively recognizable with experience. (b) The other hand requires a great deal of attention to understand the first steps are to recognize patterns within that figure.

The first identifies the major elements of the image, with the second identifying the various relationships between these. With attentive effort, the symbols are perceived (transferred from long-term memory). Patterns, mostly subsets of groups or information having perceptual or cognitive commonality, are extracted from the overall image. The last step is identifying the most interesting things (such as the most interesting point clusters, genes, countries, or products), that is, those having the most interesting or special features.

Important: Without external (cognitive) identification, a graphic is unusable. The external identification must be directly readable and understandable. Since much of our perception is driven by physical interpretations, meaningful images must have easily interpretable x, y and z-dimensions and the graphics elements of the image must be clear.

ii. Features of Graphics

Graphics have three (or more) dimensions. a matrix with points of various sizes within each cell. Every point of the graphic can be interpreted as a relation between a position in x and a position in y. The points vary in size providing a third dimension or variable to interpret. In effect, this can be considered a value in z. This produces a one-to-one correspondence between a 3D view with height and a 2D view with size, thus different interpretations for the z value. Several relationships pop out, such as the size of the circles along the diagonal (the circles in the upper right are larger than those in the lower left). The set of all points either in the 2D or 3D image represents the totality of the relations among the three dimensions x, y, and z, and any patterns present imply a pattern in the data.



Rules of a graphic All graphics are represented on the screen. All objects will be interpreted as flat (in 2D) or as physical objects (in 3D). As we saw in Chapter 3, any perceptual interpretation of the data will assume that the graphic represents parts of a 3D scene. So 3D is the medium by which we need to interpret the graphic.

Analysis of a graphic Analyzing a graphic we first perceive groups of objects. Attempt to characterize these groups (cognitively). Finally we examine special cases not within the groups or relationships between the groups (combination of both). This process can be done at many levels and with many different visualizations. Supporting analysis plays a significant role (for example, we can cluster the data and show the results of the computation, hence speeding up the likely perception of groups).

2.1.3 The Eight Visual Variables:

The application of graphics to communicate information requires an understanding of graphic primitives and their properties. All graphic primitives will be termed marks. One way to encode data for display is to map different data values to different marks and their attributes. However marks by themselves do not define informative displays since all the marks would simply obscure all previously drawn marks it is only through the spatial arrangement of marks that informative displays are created. The positioning of marks within a display space provides a means to map or reveal additional properties of the underlying data including similarity and distributions. Once the layout and types of marks are specified then additional graphical properties can be applied to each mark.

i. Position: The first and most important visual variable is that of position, the placement of representative graphics within some display space, be it one-, two-, or three-dimensional. Position has the greatest impact on the display of information, because the spatial arrangement of graphics is the first step in reading a visualization. In essence, the maximization of the spread of representational graphics throughout the display space maximizes the amount of

information communicated, to some degree. The visualization display with the worst case positioning scheme maps all graphics to the exact same position; consequently, only the last-drawn graphic is seen, and little information is exchanged. The best positioning scheme maps each graphic to unique positions, such that all the graphics can be seen with no overlaps.

ii. Mark: The second visual variable is the mark or shape: points, lines, areas, volumes, and their compositions. Marks are graphic primitives that represent data. Any graphical object can be used as a mark including symbols, letters, and words. When working purely with marks it is important not to consider differences in sizes, shades of intensity, or orientation, as these are additional visual variables that will be described later.

iii. Size (Length, Area, and Volume): The previous two visual variables, position and marks, are required to define visualization. Without these two variables there would not be much to see. The remaining visual variables affect the way individual representations are displayed these are the graphical properties of marks other than their shape. The third visual variable and first graphic property is size. Size determines how small or large a mark will be drawn. Size easily maps to interval and continuous data variables, because that property supports gradual increments over some range.

iv. Brightness: The fourth visual variable is brightness or luminance. Brightness is the second visual variable used to modify marks to encode additional data variables. While it is possible to use the complete numerical range of brightness values as human perception cannot distinguish between all pairs of brightness values.

v. Color: The fifth visual variable is color; see Chapter 3 for a detailed discussion of color and of how humans perceive color. While brightness affects how white or black colors are displayed, it is not actually color. Color can be defined by the two parameters, hue and saturation. Hue provides what most think of as color the dominant wavelength.

vi. Orientation: The sixth visual variable is orientation or direction. Orientation is a principal graphic component behind iconographic stick figure displays. Some common color maps standard linear grayscale, rainbow, heated, blue to cyan, and blue to yellow.

vii. Texture: The seventh visual variable is texture. Texture can be considered as a combination of many of the other visual variables including marks (texture elements), color (associated with each pixel in a texture region), and orientation (conveyed by changes in the local color). Dashed and dotted lines, which constitute some of the textures of linear features can be readily differentiated as long as only a modest number of distinct types exist. Varying the color of the segments or dots can also be perceived as a texture.

viii. Motion: The eighth visual variable is motion. In fact, motion can be associated with any of the other visual variables, since the way a variable changes over time can convey more information. One common use of motion is in varying the speed at which a change is occurring (such as position change or flashing, which can be seen as changing the opacity). The eye will be drawn to graphical entities based not only on similarities in behavior, but also on outliers. The other aspect of motion is in the direction; for position, this can be up, down, left, right, diagonal, or basically any slope, while for other variables it can be larger/smaller, brighter/dimmer, steeper/shallower angles, and so on.

2.1.4 Historical Perspective:

The various visualization techniques can be found, we still lack a comprehensive language to describe our graphical creations. The art of visualization the principles of graphics and their comprehension is generally understood. But as a science we have yet to define a consistent formalism for general visualizations or even just for the class of data visualizations. Researchers are now starting to look into such an idea through the development of various models Robertson first proposed this need for formal models as a foundation for visualization systems.

i. Bertin (1967) Semiology of Graphics: Jacques Bertin possibly the most important figure in visualization theory, published his *Sémiologie Graphique*. This was the first rigorous attempt at defining graphics and its application to the portrayal of information. Bertin presents the fundamentals of information encoding via graphic representations as a semiology a science dealing with sign systems. His first key point is the strict separation of content (the information to encode) from the container (the properties of the graphic system). To fully comprehend a sign system one must first completely understand the primitive elements that define such a system. Consequently Bertin embarks on defining a graphical vocabulary.

ii. Mackinlay (1986) APT Mackinlay introduced a design for an automated graphical presentation designer of relational information, named APT (A Presentation Tool). APT was designed to extract some information from a database and to render a graphical design that presented this information. A graphical design is an abstract description of the graphical techniques encoding information. Like Bertin, Mackinlay focused his research on static two-dimensional presentations, such as bar charts, scatterplots, and connected graphs however, Mackinlay went on to describe graphical languages, defining graphical presentations as sentences of these languages.

A graphical sentence s is defined to be a collection of tuples: $s \subset \{ \langle o, l \rangle : o \in O \wedge l \in L \}$, where O is a set of graphical objects and L is a set of locations. Each tuple which is called a located object indicates the placement of an object at a given location. The syntax of a graphical language is defined to be a set of well-formed graphical sentences.

iii. Bergeron and Grinstein (1989) Visualization Reference Model the Visualization Reference Model by Bergeron and Grinstein defines an abstraction of the visualization problem, which establishes a mapping from the underlying data space to a physical representation. Based on the conventional graphics system's viewing pipeline, the model is represented by a conceptual visualization pipeline. This pipeline is organized into four stages. The first stage identifies the source and provides appropriate information about the data structure. Standardized data from the previous stage enters the model transformation stage which defines appropriate projections of the source data space to a usable representation data space.

iv. Wehrend and Lewis (1990) Wehrend and Lewis also defined a mechanism for automatically defining visualizations. They constructed a large catalog of encoding techniques and their effective uses. The catalog is arranged as a two-dimensional matrix classified as objects and operations. The objects identify the problems and are grouped together based on their target domains, while the operations identify groups of similar goals. The catalog is filled with problems (tasks to perform) and solutions (visualization techniques that provide answers). The resulting catalog matches domain tasks and desired goals with appropriate visualization techniques.

v. Robertson (1990) Natural Scene Paradigm The Natural Scene Paradigm introduced by Robertson aims to visually display data represented by identifiable properties of realistic scenes. Robertson reasoned that people have highly developed skills for analyzing multiple aspects of natural scenes, and aimed to exploit these skills for multivariate analysis. Natural scene views are defined as two- or three-dimensional spatial surfaces with spectral and temporal variables. Visual properties such as surface height, material, density, phase, and wetness are defined and ranked, based on perceptual characteristics.

vi. Roth (1991) Visage and SAGE Roth et al created Visage a prototype user-interface environment for exploring information which incorporates SAGE a knowledge-based automatic graphic design tool, and extends the ideas of Mackinlay for general two-dimensional graphics. The primary contribution of Visage is its "information-centric" approach where the central focus of user interaction is connected directly to the data elements (graphic representations). The whole environment is based on two basic object types elements and frames. The elements are the various data representations the graphic objects of visualization.

vii. Casner (1991) BOZ BOZ, developed by Stephen Casner is an automated graphic design presentation tool to assist in performing specific tasks. The main focus of BOZ is to replace logical task descriptions with perceptually equivalent ones by encoding logical inferences (mental arithmetic or numerical comparisons) with perceptual inferences (shortest distance and average size) from which solutions can be visually obtained. BOZ can be used to design different presentations of the same information customized to the requirements of different tasks.

viii. Beshers and Feiner (1992) AutoVisual AutoVisual is an automatic system for designing visualizations within the n-Vision visualization system. The n-Vision system implements the worlds-within-worlds visualization technique that recursively defines subspace coordinate systems, and is defined as a hierarchy of interactors consisting of four components: encoding objects, encoding spaces, selections, and a user interface.

ix. Senay and Ignatius (1994) VISTA Senay and Ignatius extended the work of Mackinlay but focused on scientific data visualization. They developed VISTA (Visualization Tool Assistant) a knowledge-based system for visualization design. Senay and Ignatius like Bertin and Mackinlay implement composition rules for the generation of complex visualizations

from simple visualization techniques. They start by identifying three sub-processes defining the visualization pipeline: data manipulation, visualization mapping, and rendering. These three transformations process the data into graphic representations that are rendered to a display.

x. Hibbard (1994) Lattice Model Hibbard presents a lattice model for describing visualizations. Unlike the previous graphical models that focused on the graphic primitives, the lattice model focuses on data to display transformations. Hibbard notes, “data objects are approximations to mathematical objects and real displays are approximations to ideal displays”. He considers the process of specifying visualizations as a mathematical transformation from data to displays.

xi. Golovchinsky (1995) AVE AVE (Automatic Visualization Environment) is an automatic graphical presentation system based on a generative theory of diagram design, the construction of diagrams from basic components corresponding to relations present in the data. Diagrams are composed of graphical elements only rectangles in this implementation that have attributes and are related to other elements through graphical relations based on the underlying data relations. The resulting graphics are trees and graphs depicting nodes as rectangles and relationships with lines or arrows.

xii. Card, Mackinlay, and Shneiderman (1999) Spatial Substrate Information Visualization Using Vision to Think, Card, Mackinlay, and Shneiderman summarize information visualization research and applications. Interest here is their section on “mapping data to visual form.” Within this section Card et al. present a reference model for visualizations describing three primary transformations for mapping data to visual form that also support human interaction: data transformations, visual mappings, and view transformations.

xiii. Kamps (1999) EAVE EAVE (Extended Automatic Visualization Engine) by Kamps is an extension of AVE. EAVE takes arbitrary relations as input and generates diagram visualizations. While the diagrams generated by this system primarily depend on data characteristics and graphical knowledge, user preferences are also taken into account. This system only generates traditional types of diagrams that are commonly used in publications.

ivx. Wilkinson (1999) Grammar of Graphics Wilkinson’s Grammar of Graphics based on his original Graphics Algebra specifies the construction of statistical graphics. This grammar of graphics is actually a grammar of statistical visualizations a subclass of data visualizations readily used for statistical analyses. Each individual component of these graphics is defined as instances of various graphical objects the combination of individual components defining the resulting display.

vx. Hoffman (2000) Table Visualizations Hoffman’s model was the first attempt at defining a generalized space of data visualizations. It encapsulates primitive-graphic properties and combines techniques like survey plots, scatterplots, RadViz, and parallel coordinates within geometric layouts.

2.1.5 Taxonomies:

Taxonomy is a means to convey a classification. Often hierarchical in nature taxonomy can be used to group similar objects and define relationships. In visualization we are interested in many forms of taxonomies including data visualization techniques tasks and methods for interaction. In this section we briefly describe a number of such taxonomies from the literature. Interested readers are directed to the cited papers.

i. Keller and Keller (1994) Taxonomy of Visualization Goals Keller and Keller Visual Cues classify visualization techniques based on the type of data being analyzed and the user’s task(s). Similar to those identified earlier in this book, the data types they consider are:

- scalar (or scalar field),
- nominal,
- direction (or direction field) shape,
- position,
- Spatially extended region or object (SERO).

The authors also define a number of tasks that a visualization user might be interested in performing. While some of the tasks seem interrelated, their list is a useful starting position for someone setting out to design visualization for a particular application.

- identify—establish characteristics by which an object is recognizable;
- locate—ascertain the position (absolute or relative);
- distinguish—recognize as distinct or different (identification is not needed);

- categorize—place into divisions or classes;
- cluster—group similar objects;
- rank—assign an order or position relative to other objects;
- compare—notice similarities and differences;
- associate—link or join in a relationship that may or may not be of the same type;
- Correlate—establish a direct connection, such as causal or reciprocal.

ii. Shneiderman (1996) Data Type by Task Taxonomy

A related strategy was proposed by Shneiderman. His list of data types was somewhat different from Keller and Keller's, and included more types from the information visualization field. His list of data types consisted of:

- one-dimensional linear;
- two-dimensional map;
- three-dimensional world;
- temporal;
- multidimensional;
- tree;
- network.

Shneiderman looked more at the behavior of analysts as they attempt to extract knowledge from the data. His task set consisted of the following:

- **Zoom** Zoom in items of interest to gain a more detailed view, e.g., holding down a mouse button to enlarge a region of the display.
- **Filter** Filter out uninteresting items to allow the user to reduce the size of a search, e.g., dynamic queries that can be invoked via sliders.
- **Details-on-demand** Select an item or group and get details when needed, e.g., a pop-up window can show more attributes of a specific object on the screen.
- **Relate** View relationships among items, e.g., select a particular object that can then show all other objects related to it.
- **History** Keep a history to allow undo, replay, and progressive refinement, such as allowing a mistake to be undone, or a series of steps to be replayed.
- **Extract** Extract the items or data in a format that would facilitate other uses, i.e., saving to file, sending via e-mail, printing, or dragging into another application (statistical or presentation package).

iii. Keim (2002) Information Visualization Classification

Keim designed a classification scheme for visualization systems based on three dimensions: data types, visualization techniques, and interaction/distortion methods. His interaction/distortion technique classification has some similarities to Shneiderman's tasks, as do his data types, but his classification of the visualization techniques used is not included in the other taxonomies. The components of each of his classification dimensions are listed below:

- **Classification of Data Types.** 6 types of data exist:
 - One-dimensional data—e.g., temporal data, news data, stock prices, text documents
 - Two-dimensional data—e.g., maps, charts, floor plans, newspaper layouts
 - Multidimensional data—e.g., spreadsheets, relational tables
 - Text and hypertext—e.g., new articles, web documents
 - Hierarchies and graphs — telephone/network traffic, system dynamics models
 - Algorithm and software — software, execution traces, memory dumps
- **Classification of Visualization Techniques.** 5 classes of visualization techniques exist:
 - Standard 2D/3D displays—e.g., x, y- or x, y, z-plots, bar charts, line graphs;
 - Geometrically transformed displays—e.g., landscapes, scatterplot matrices, projection pursuit techniques, projection views, hyperslice, parallel coordinates;
 - Iconic displays—e.g., Chernoff faces, needle icons, star icons, stick figure icons, color icons, tilebars;
 - Dense pixel displays—e.g., recursive pattern, circle segments, graph sketches;
 - Stacked displays—e.g., dimensional stacking, hierarchical axes, worlds-within-worlds, treemaps, cone trees.
- **Classification of Interaction and Distortion Techniques.** 5 classes of interaction techniques exist:
 - Dynamic projection—e.g., grand tour system, XGobi, XLispStat, ExplorN;

- Interactive filtering—e.g., Magic Lenses, InfoCrystal, dynamic queries, Polaris;
- Interactive zooming—e.g., TableLens, PAD++, IVEE/Spotfire, DataSpace, MGVI and scalable framework;
- Interactive distortion—e.g., hyperbolic and spherical distortions, bifocal displays, perspective wall, graphical fisheye views, hyperbolic visualization, hyperbox;
- Interactive linking and brushing—e.g., multiple scatterplots, bar charts, parallel coordinates, pixel displays and maps, Polaris, scalable framework, S-Plus, XGobi, XmdvTool, DataDesk.

Experimental semantics based on perception: is a cross-disciplinary field investigating how humans construct and process linguistic meaning by anchoring words to sensory, motor, and affective experiences. Rather than treating language as abstract symbols, it uses behavioral and neuroscientific experiments to show that mental representations of words rely on re-creating actual perceptual.

- **Simulation Semantics:** Words like "kick" or "grasp" are processed by mentally simulating the corresponding physical action. Brain imaging shows that understanding these verbs instantly activates the exact motor and sensory cortices responsible for performing them.
- **Perceptual Effects on Word Recognition:** Semantic knowledge directly influences low-level perception. Studies demonstrate that being provided with the function of unfamiliar objects immediately alters visual cortical processing in under 200ms.
- **Multimodal Integration:** Meaning isn't just visual or auditory; it spans the full sensorimotor spectrum. Researchers study how tactile experiences, spatial orientation, and even social cues like eye contact and blinking establish meaning in interpersonal perception.
- **Methodologies:** The discipline relies on controlled empirical experiments using methods like eye-tracking, reaction-time tests, and functional neuroimaging (fMRI/MEG) to observe semantic processing.

2.2 Gibson's Affordance Theory:

According to Gibson, affordances are physical properties of the environment that we directly perceive. Many theorists, unlike Gibson, think of perception as a very active process: The brain deduces certain things about the environment based on the available sensory evidence.

Limitations:

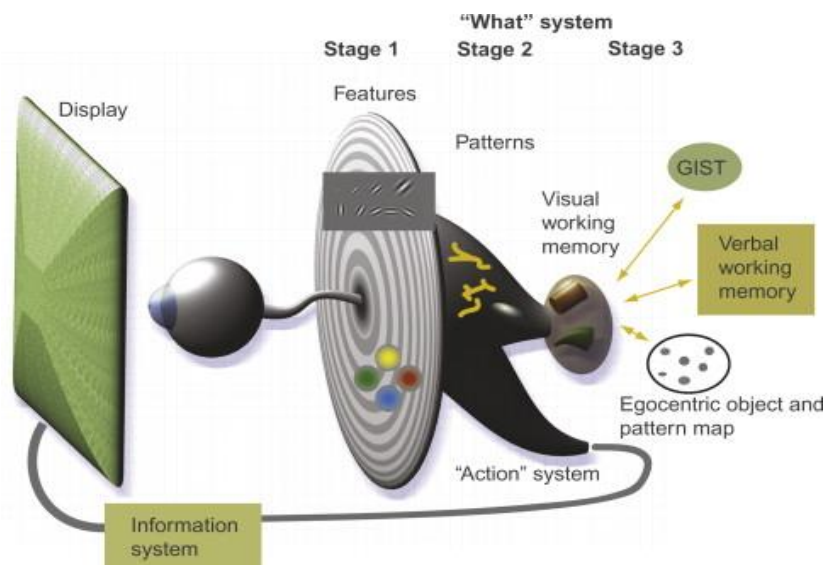
There are three problems with Gibson's direct perception in developing a theory of visualization.

- The first problem is that even if perception of the environment is direct it is clear that visualization of data through computer graphics is very indirect. Many layers of processing between the data and its representation which are invisible
- Second, there are no clear physical affordances in any graphical user interface.
 - To say that a screen button "affords" pressing in the same way as a flat surface affords walking
 - the use of buttons is arbitrary-when pressed do interesting things which are indirect
- Third, Gibson's rejection of visual mechanisms is a problem
 - To reject the importance of understanding visual mechanisms would be to reject a tremendous proportion of vision research as irrelevant

A Model of Perceptual Processing:

There are many subsystems in vision, and we should always be wary of overgeneralization. Still, an overall conceptual framework is useful in providing a starting point for more detailed analysis. A broad schematic overview of a three-stage model of perception.

- In Stage 1, information is processed in parallel to extract basic features of the environment.
- In Stage 2, active processes of pattern perception pull out structures and segment



the visual scene into regions of different color, texture, and motion patterns.

- In Stage 3, the information is reduced to only a few objects held in visual working memory by active mechanisms of attention to form the basis of visual thinking.

Stage 1: Parallel Processing to Extract Low-Level Properties of the Visual Scene

- Visual information is first processed by large arrays of neurons in the eye and in the primary visual cortex at the back of the brain.
- Selective tuning of information for a particular kind of information such as the orientation of edges or the color of a patch of light.
- Important characteristics of Stage 1 processing include:
 - Rapid parallel processing
 - Extraction of features, orientation, color, texture, and movement patterns
 - Transitory nature of information, which is briefly held in an iconic store
 - Bottom-up, data-driven model of processing

Stage 2: Pattern Perception

- At the second stage, rapid active processes divide the visual field into regions and simple patterns.
- The pattern-finding stage of visual processing is extremely flexible, influenced both by the massive amount of information available from Stage 1 parallel processing and by the top-down action of attention driven by visual queries.
- Marr (1982) called this stage of processing the 2-1/2D sketch.
- Triesman (1985) called it a feature map.
- Rensink (2002) called it a proto-object flux to emphasize its dynamic nature.
- There is increasing evidence that tasks involving eye-hand coordination and locomotion may be processed in pathways distinct from those involved in object recognition.
- This is the two-visual system hypothesis: one system for locomotion and action, called the "action system," and another for symbolic object manipulation, called the "what system."
- Important characteristics of Stage 2 processing include:
 - Slow serial processing Involvement of both working memory and long-term memory
 - More emphasis on arbitrary aspects of symbols
 - In a state of flux, a combination of bottom-up feature processing and top-down attentional mechanisms
 - Different pathways for object recognition and visually guided motion

Stage 3: Sequential Goal-Directed Processing

- At the highest level of perception are the objects held in visual working memory by the demands of active attention.
- At this level, only a few objects can be held at a time; they are constructed from the available patterns providing answers to the visual queries.
- The visual object identification process interfaces with the verbal linguistic subsystems of the brain so that words can be connected to images.
- The perception-for-action subsystem interfaces with the motor systems that control muscle movements.